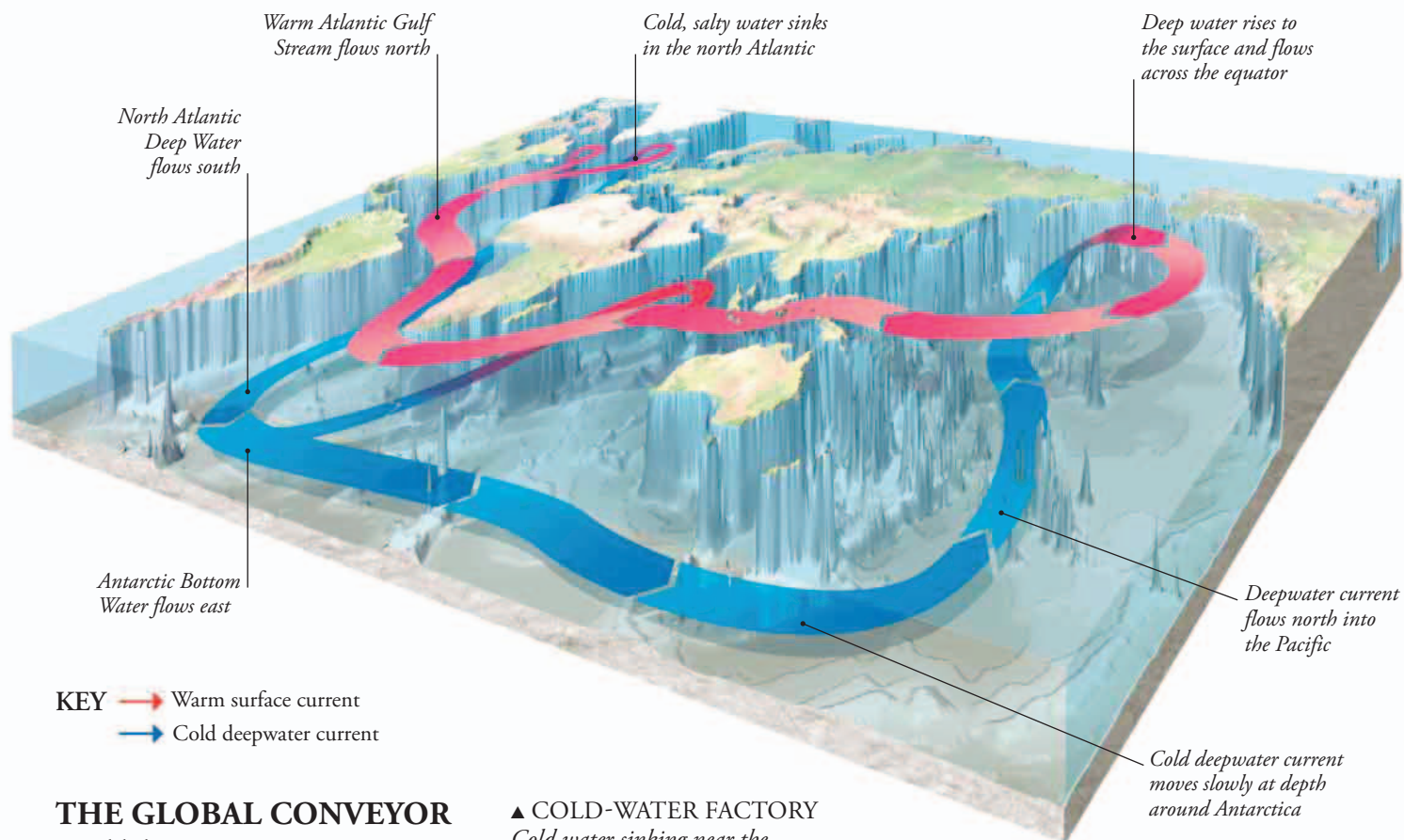




THE GLOBAL CONVEYOR

A cold deepwater current sweeps through the Southern Ocean and into the Indian and Pacific oceans. Here, some of it rises to feed into the surface currents. These link with the warm Atlantic Gulf Stream, which eventually cools and sinks in the far north to drive the flow. The whole network is often called the global conveyor, because it conveys ocean water all around the globe.

▲ COLD-WATER FACTORY

Cold water sinking near the poles is the main driving force behind this never-ending circulation pattern.

FAST FACTS

- Scientists describe the global conveyor as the thermohaline circulation, driven by heat (*thermos*) and saltiness (*haline*).
- It takes about 1,000 years for a single drop of ocean water to travel all around the world in the global conveyor.
- Deepwater currents flow faster where they squeeze between the continents.

ESSENTIAL SUPPLIES

The global conveyor carries ocean water around the world, along with dissolved oxygen and nutrients that are vital to oceanic life. A lot of these nutrients are scoured from the ocean floor by the deepwater currents, which eventually carry them up to the sunlit surface. Here, they nourish the plankton that feed animals like these humpback whales.



SLOWDOWN

The world's climate is changing, and this could affect the global conveyor. Global warming is melting Arctic ice, adding freshwater to the sea. This is making the North Atlantic seawater less salty and less likely to sink and drive deepwater currents. Since the sinking water draws the warm Gulf Stream north toward Europe, this could lead to Europe getting colder.